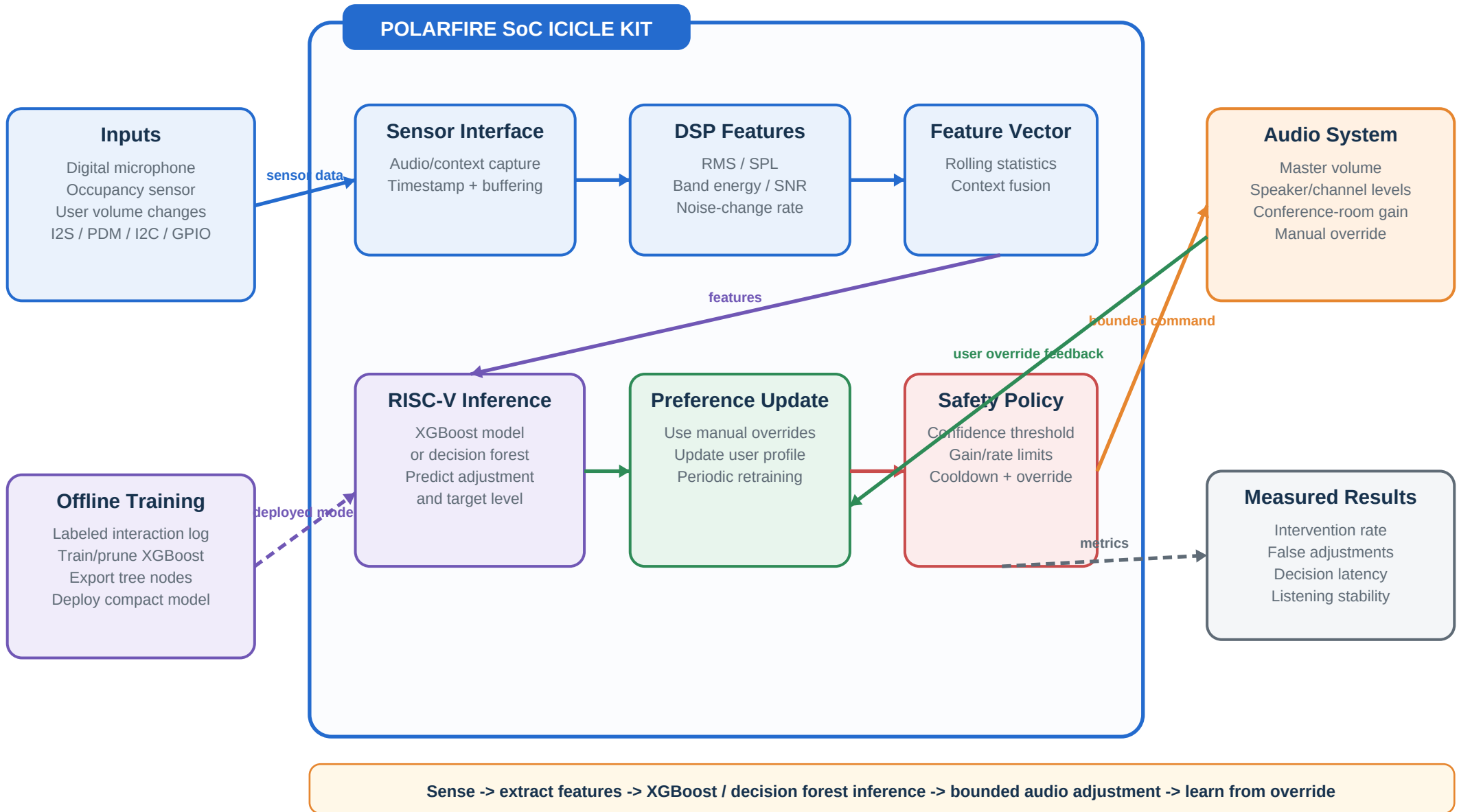


Predictive Acoustic Comfort - Simplified System Architecture

PolarFire SoC Icicle Kit: FPGA feature pipeline + RISC-V preference model + closed-loop audio control



Model choice: start with XGBoost for training accuracy; deploy a pruned tree ensemble as compact C/C++ inference on RISC-V.
FPGA fabric handles deterministic streaming DSP. RISC-V handles inference, preference profiles, control policy, and model updates.